THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In "If a number is divisible by 10, then its ones digit is 0", the hypothesis is: 1

(a) its ones digit is 0
(b) it is not true that a number is divisible by 10
(c) a number is divisible by 10, then its ones digit is 0
(d) a number is divisible by 10

2. In "If a number is divisible by 10, then its ones digit is 0", the conclusion is: 1

(a) a number is divisible by 10
(b) its ones digit is 0
(c) a number is divisible by 10, then its ones digit is 0
(d) it is not true that its ones digit is 0

3. The statement "Every prime number is odd" is: 1

(a) true only for prime numbers **(b)** true only for even numbers
(c) false **(d)** true

4. "Every square is a rectangle" can be written in if–then form as: 1

(a) A figure is a square or it is a rectangle.
(b) If it is a rectangle, then a figure is a square.
(c) If not (a figure is a square), then it is a rectangle.
(d) If a figure is a square, then it is a rectangle.

5. The number of factors of 9 is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

6. The converse of the proposition "If a number is a multiple of 8, then it is a multiple of 4" is: 1

(a) If it is not true that a number is a multiple of 8, then it is not true that it is a multiple of 4.
(b) If a number is a multiple of 8, then it is a multiple of 4.
(c) If a number is a multiple of 4, then it is a multiple of 8.
(d) If it is not true that it is a multiple of 4, then a number is a multiple of 8.

7. Which of the following numbers is divisible by 9? 1

(a) 730 **(b)** 731 **(c)** 729 **(d)** 725

8. The proposition "If a number is divisible by 10, then its ones digit is 0" is true. Its converse "If its ones digit is 0, then a number is divisible by 10" is: 1

(a) True **(b)** False
(c) True only for some values **(d)** It cannot be decided

9. In "If two lines are parallel, then the alternate interior angles made by a transversal are equal", the hypothesis is: 1

(a) it is not true that two lines are parallel
(b) two lines are parallel, then the alternate interior angles made by a transversal are equal
(c) the alternate interior angles made by a transversal are equal
(d) two lines are parallel

10. In "If a quadrilateral is a square, then it is a rectangle", the conclusion is: 1

(a) it is a rectangle
(b) a quadrilateral is a square, then it is a rectangle
(c) it is not true that it is a rectangle
(d) a quadrilateral is a square

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the converse of "If a number is divisible by 10, then its ones digit is 0". Is it true? 2

12. Write "A square has four equal sides" in if–then form and name the hypothesis. 2

13. Find the number of factors of 9. Is 9 a perfect square? 2

14. Write the converse of "If two lines are parallel, then the alternate interior angles made by a transversal are equal". Is it true? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of "If a quadrilateral is a rhombus, then its diagonals are perpendicular". Is the converse true? Justify. 3

16. Show that the statement "If $x^2 > 9$, then $x > 3$" is false by giving a counterexample. 3

17. Is a triangle with sides 6 cm, 8 cm and 10 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Write the converse of "If a number is a multiple of 4, then it is even". Is the converse true? Justify. 3

18. Show that the statement "If a number is divisible by 4 and by 6, then it is divisible by 24" is false by giving a counterexample. 3

OR

Is a triangle with sides 5 cm, 12 cm and 14 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Verify with 3 and 5 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 6 and 7. 5

20. Verify with 7 and 11 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 9 and 10. 5

END OF QUESTION PAPER — **Total: 40 Marks**